

Report
Of
VIRTUAL INTERNSHIP AT SKYSCANNER

Submitted
To
School of Computer Science and Engineering
IIIM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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2026-27

CANDIDATE'S DECLARATION

I, Akarsh Anand, do hereby declare that the internship report titled "Virtual Internship at Skyscanner (Forage Program)" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and I shall be solely responsible for any kind of copyright violation in this regard.

A handwritten signature in blue ink that reads "Akarsh Anand". The signature is written in a cursive style with a large initial 'A'.

Signature

Student Name: Akarsh Anand

Roll No.: 25SCS1003000760

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to Skyscanner and Forage for providing this virtual internship opportunity which gave me practical exposure to real-world software development tasks in the travel-technology domain. I am also thankful to my faculty mentors at IILM University for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

A handwritten signature in blue ink that reads "Akarsh Anand". The signature is written in a cursive style with a large initial 'A'.

Signature

Student Name: Akarsh Anand

Roll No.: 25SCS1003000760

INTERNSHIP COMPLETION CERTIFICATE

Skyscanner



Inspiring and
empowering
future professionals

Akarsh Anand Software Engineering Job Simulation

Certificate of Completion
August 26th, 2026

Over the period of August 2026, Akarsh Anand has completed practical tasks in:

Create a Backpack React web app
Create a microservice
Create a mobile flight itinerary
Build a user-friendly platform

A handwritten signature in black ink, appearing to read "Tom Brunskill".

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a8f51507222428647374d4d | User Verification Code 6a8f4dafd8fa146772b3119f | Issued by Forage

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4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work completed during a virtual internship program offered by Skyscanner through the Forage platform. Forage partners with leading global companies to provide students with practical, industry-relevant, job-simulation tasks that closely mirror the actual work carried out by employees of the organization. The Skyscanner Software Engineering Virtual Internship consists of four introductory-level modules, each designed to be completed in one to two hours, covering front-end web development, backend microservices, mobile application development, and UI/UX design. The four tasks undertaken as part of this internship were: (1) Creating a Backpack React web application, (2) Creating a Dropwizard microservice for car rentals and hotels, (3) Creating a mobile flight itinerary application, and (4) Building a user-friendly platform through UX/UI design. Together, these tasks provided a well-rounded, full-stack perspective on how Skyscanner builds consistent, scalable, and user-friendly travel technology products.

4.2 Organization Profile

Skyscanner is a global travel search engine and metasearch platform headquartered in Edinburgh, Scotland, that enables users to compare flights, hotels, and car rental prices across hundreds of providers. Founded in 2003, Skyscanner has grown into one of the most widely used travel platforms in the world, serving millions of users across different countries and languages. The company is known for its strong engineering culture, its investment in design systems such as Backpack, and its data-driven approach to building reliable, high-performance travel products. Skyscanner's engineering teams work across web, mobile, and backend platforms to ensure a consistent and seamless experience for users searching, comparing, and booking travel options.

4.3 Problem Statement

Large-scale travel platforms such as Skyscanner face the challenge of delivering a consistent user experience across multiple platforms (web and mobile) while supporting a wide range of backend services such as flight

search, hotel booking, and car rental comparison. Inconsistent design components, fragmented backend microservices, and a lack of well-designed, intuitive interfaces can lead to a poor and confusing user experience. The four tasks of this internship were framed around solving these real challenges: building reusable UI components using a shared design system (Backpack), creating an independent backend microservice to serve car rental and hotel data, developing a mobile interface to present flight itinerary information clearly, and designing an easy-to-use platform interface that developers and designers can implement consistently.

4.4 Project Objectives

To understand and apply Skyscanner's Backpack design system to build a functional React web application.

To design and implement a Dropwizard-based backend microservice capable of assisting users in finding car rentals and hotels.

To develop a basic Android mobile application that displays a user's flight itinerary in a clear and structured format using Backpack components.

To create user-centric design mockups/wireframes that improve ease of use for both end users and the development team.

To gain hands-on exposure to real-world, industry-style software engineering tasks spanning front-end, backend, mobile, and UX/UI design.

4.5 Scope of the Project

The scope of this internship was limited to four introductory, self-contained job-simulation tasks provided by Forage on behalf of Skyscanner. Each task was independent in nature but collectively covered the major layers of a typical travel-technology product: presentation layer (React web app), backend services layer (Dropwizard microservice), mobile application layer (Android app), and design layer (UI/UX platform design). The project did not involve integration with Skyscanner's live production systems or real user data; all tasks were completed in a simulated, sandboxed learning environment intended for skill-building and portfolio development.

4.6 Technologies and Tools Used

Frontend: React.js, Skyscanner Backpack (BPK) React component library, JavaScript/J SX, HTML5, CSS3.

Backend: Java, Dropwizard framework, RESTful APIs, Maven.

Mobile: Android Studio, Kotlin/Java, Backpack for Android (BPK Android components), XML layouts.

Design: Figma (wireframing and prototyping), Skyscanner Backpack design tokens, UX research and usability principles.

Others: Git/GitHub for version control, Forge virtual lab environment for task simulation.

4.7 System Architecture

The overall system architecture followed during this internship reflects a typical layered travel-application architecture. At the presentation layer, the React web application and the Android mobile application consume shared design components from the Backpack library ensuring visual and functional consistency across platforms. At the service layer, the Dropwizard microservice exposes RESTful endpoints that can be queried by client applications to retrieve car rental and hotel information following a standard client-server request/response model. The UX/UI design layer sits above these technical layers, defining the wireframes, component usage guidelines, and interaction patterns that both the web and mobile applications are built to follow, ensuring a unified user experience across the platform.

4.8 Methodology

Each task was approached using a structured, iterative methodology consistent with standard software engineering practice:

Task 1 (Backpack React Web App): Set up a React development environment, installed the Backpack React component library, and used pre-built components (buttons, cards, navigation bars) to assemble a basic, styled web page consistent with Skyscanner's design language.

Task 2 (Dropwizard Microservice): Configured a Dropwizard project,

defined REST resource classes and endpoints, implemented basic business logic to return car rental and hotel search results, and tested the endpoints locally.

Task 3 (Mobile Flight Itinerary App): Created a new Android Studio project, integrated Backpack for Android components, and designed screens/layouts to display flight itinerary details such as departure/arrival times, airports, and flight duration.

Task 4 (UX/UI Platform Design): Analyzed usability requirements from a developer's and end-user's perspective, created wireframes/mockups, and applied design consistency principles to propose an easy-to-use platform layout.

4.9 Expected Outcomes

A functional understanding of building React applications using an established component/design system (Backpack).

A working backend microservice capable of serving car rental and hotel-related data via REST APIs.

A basic Android application capable of displaying structured flight itinerary information to users.

A set of design artifacts demonstrating improved usability and consistency for both developers and end users.

Overall, a practical, portfolio-ready demonstration of full-stack, cross-platform software engineering skills gained through the Skyscanner Forage virtual internship.

5. BIBLIOGRAPHY/REFERENCES

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